

MetaGPT: Enable Reliable Multi-Agent Software Delivery

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Project: github.com/geekan/MetaGPT

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Stabilize Collaboration With Structured SOP Artifacts

- LLM multi-agent systems suffer cascading hallucinations and incoherence.
- MetaGPT encodes SOPs into roles and prompt sequences with structured handoffs.
- Agents produce PRDs, designs, flowcharts, and interface specs to reduce errors.

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Figure 1: Software development SOPs mirrored by MetaGPT roles and workflows.

Boost Executability And Cut Human Revisions

- Pass@1: HumanEval 85.9%, MBPP 87.7% (MetaGPT + GPT-4).
- SoftwareDev: Executability 3.75 vs. ChatDev 2.25; Human revisions 0.83 vs. 2.5.

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Figure 4: MBPP & HumanEval pass rates.

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Figure 5: Demo software generated by MetaGPT.

Drive Coherence With Standardized Intermediate Outputs

- SOPs ensure task decomposition, role clarity, and standards for outputs.
- MetaGPT leverages meta-programming via specialized agents and structured deliverables.

Prefer Structure Over Free-Form Agent Dialogue

Automatic Programming

- From PROW (1969) to modern code LLMs (CodeBERT, CodeLlama, WizardCoder).
- ReAct / Reflexion: reasoning–action loops; tool use (ToolFormer, ToolLLM).
- Role-play alone lacks structured outputs for complex software engineering.

LLM Multi-Agent Frameworks

- Generative Agents, NLSOM, planning/strategy, LLM economies.
- Issues: repetition, loops, inconsistent collaboration; SOPs mitigate these.

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Figure 2: Communication protocol and executable feedback loop.

Enable Reliable Collaboration With SOP-Driven Roles

- Meta-programming framework for LLM-based multi-agent systems.
- Core pillars: role specialization, SOP-driven workflow, structured communication.
- Shared message pool with publish–subscribe for targeted information flow.
- Executable feedback: runtime execution, testing, and iterative repair.

Standardize Handoffs To Reduce Collaboration Errors

- Roles: Product Manager, Architect, Project Manager, Engineer, QA Engineer.
- Each role has profile, goals, constraints, tools (e.g., web search, code execution).
- Agents monitor a message pool; ReAct-style behavior for observe–think–act.
- Outputs: PRD, file lists, data structures, interface definitions, test cases.

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Figure 3: SOP-driven software development pipeline in MetaGPT.

Sequence Roles To Deliver Validated Software

- Product Manager: analyzes requirements → structured PRD (User Stories, Requirement Pool).
- Architect: system design → file list, data structures, APIs, sequence flows.
- Project Manager: task distribution per design.
- Engineer: implements; QA: test cases and quality gates.
- Final: assembled, validated software solution.

Replace Brittle Chat With Structured Artifacts

- Structured schemas per role minimize ambiguity vs. free-form chat.
- Documents and diagrams as primary artifacts for handoffs.
- Global message pool: publish once, subscribe by interest/dependencies.
- Activation when prerequisites are satisfied; reduced info overload.

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Figure 2 (left): Shared message pool and subscriptions.

Correct Runtime Errors With Executable Feedback

- Engineer executes generated code and unit tests; inspects runtime errors.
- Cross-checks against PRD and design; iteratively debugs (≤ 3 retries).
- Addresses hallucination-induced review misses; boosts executability.

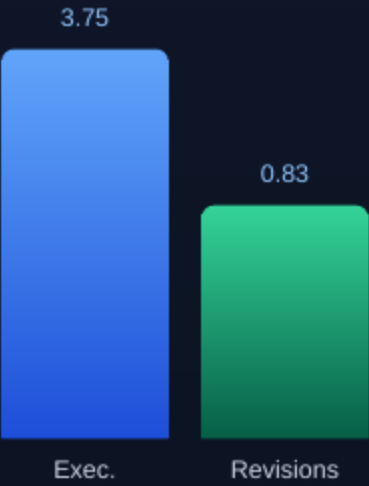
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Figure 2 (right): Code run, test, diagnose, and fix loop.

Validate On Standard And Realistic Benchmarks

- Datasets: HumanEval (Chen et al., 2021a), MBPP (Austin et al., 2021), SoftwareDev (70 tasks; 7 sampled).
- Metrics: Pass@k; executability (1–4), cost (time/tokens/\$), code stats, productivity, human revision cost.
- Baselines: AlphaCode (Li et al., 2022), Incoder (Fried et al., 2022), CodeGeeX (Zheng et al., 2023), CodeGen (Nijkamp et al., 2023), Codex (Chen et al., 2021a), CodeT (Chen et al., 2022), PaLM (Chowdhery et al., 2022), GPT-4 (OpenAI, 2023); frameworks: AutoGPT (Torantulino et al., 2023), LangChain+REPL (Chase, 2022), AgentVerse (Chen et al., 2023), ChatDev (Qian et al., 2023).

Outperform On Practical SoftwareDev Evaluations

- Executability: 3.75 (MetaGPT) vs. 2.25 (ChatDev).
- Human revisions: 0.83 (MetaGPT) vs. 2.5 (ChatDev).



Cover The End-To-End Software Lifecycle

- MetaGPT provides PRD, technical design, API generation, pre-compilation execution, role-based management, code review.
- Others: partial support; often lack structured workflows.

Summary
MetaGPT covers end-to-end SE lifecycle with SOPs and feedback.

Direct Attention With Publish–Subscribe Filters

Schemas

- Role-specific templates: PRD, API spec, sequence diagram, test plan.
- Completeness and formatting enforced at each handoff.

Publish–Subscribe

- Agents subscribe by interests/dependencies.
- Activation only when prerequisites met; less chatter.

Prevents telephone-game distortions; centers collaboration on essential artifacts.

Guide Implementation With SOP Artifacts

PRD
User stories, requirement pool, acceptance criteria.

System Design
File list, data models, API/interface specs.

Sequence Flow
Role interactions and module call sequences.

Iterate Quickly With Test-Driven Repair

```
for attempt in range(<=3): run_unit_tests() if tests_pass: ship() break else: analyze_errors()
compare(PRD, design, code) patch_code()
```

- Grounds fixes in artifacts; limits retries to ensure progress.

MetaGPT Reliably Delivers Executable Multi-Agent Software

- Structured artifacts + pub/sub + executable feedback → coherent, robust code.
- SoTA on HumanEval/MBPP; strong performance on complex software tasks.
- Promising path to regulated, human-inspired multi-agent systems.

Our Benchmarks And Baselines Ensure Fair, Reproducible Comparisons

- Use widely adopted code benchmarks (HumanEval, MBPP) with standard Pass@k to align with prior work.
- Add SoftwareDev (70 tasks; 7 sampled) to test realistic engineering metrics: executability, cost, and revision effort.
- Compare against established code models and agent frameworks to contextualize gains under common settings.

Full bibliography moved to appendix.

The Project Is Open-Sourced For Community Adoption

- Try MetaGPT, reproduce results, and extend SOP-driven agents in your workflows:
<https://github.com/geekan/MetaGPT>